

# Negative Exponents with Variable Bases

Teacher's Edition

## The Period

**MA.8.AR.1.1** Apply the Laws of Exponents to generate equivalent algebraic expressions, limited to integer exponents and monomial bases.

**Must** Integer exponents on monomial bases, negatives included: rewrite so that only positive exponents are left, and combine several laws in one expression.

**Must not** Variables are nonzero wherever a negative exponent appears. No rational exponents, and monomials only.

**Target** I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

**MTR MTR.5.1** — Notes I · **MTR.6.1** — Board 4 · **MTR.4.1** — Board 6

| Segment                             | Slides | Time          |
|-------------------------------------|--------|---------------|
| Settle in; post the learning target | 1      | 1 min         |
| Warm-Up                             | 2-3    | 5 min         |
| Notes                               | 4-6    | 11 min        |
| Example 1                           | 7-8    | 3 min         |
| Your Turn                           | 9-10   | 3 min         |
| Example 2                           | 11-12  | 3 min         |
| Your Turn                           | 13-14  | 3 min         |
| Whiteboards                         | 15-32  | 13 min        |
| Independent Set                     | 33     | 6 min         |
| IXL                                 | 34     | 5 min         |
| <b>Total</b>                        |        | <b>53 min</b> |

### Say these three things out loud today

1. A negative exponent on a letter means what it meant on a number: reciprocal.
2. Only the factor wearing the negative exponent moves. In  $3x^{-2}$ , the 3 stays upstairs.
3. Simplified means no negative exponents are left. Check every one before the board goes up.

## Slide by Slide

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1. **Negative Exponents with Variable Bases.** Post the learning target.
2. **Warm-Up — On your own. Four minutes. No notes.** Q2 and Q4 together are the lesson: a negative exponent asks for a reciprocal.
3. **Warm-Up — Answers.** Reveal. 1.  $27x^6$  2.  $1/25$  3. 9 4.  $\frac{3}{2}$
4. **Notes — Copy the table.**  $x^{-3}$  is one over  $x^3$ . It is a reciprocal. It is not negative, and it is not  $-3x$ .
5. **Notes.** Only the factor wearing the negative exponent moves. Everything else stays put.
6. **Notes.** Product and quotient laws are unchanged; the exponents are integers now, so the sums and differences can land below zero.
7. **Example 1 — Multiplying monomials with negative exponents.** SPOKEN ONLY.
8. **Example 1 — Worked.** Add the exponents of each variable, then move the  $y^{-2}$  down.
9. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the coefficient-mover ( $a^3/(4b^2)$ ) and the sign-dropper ( $4a^7b^4$ ).
10. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
11. **Example 2 — Several laws in one expression — the state guide's item.** SPOKEN ONLY.
12. **Example 2 — Worked.** Raise every factor to the 5th:  $a^{15}$  on top,  $4^5 = 1,024$  and  $b^{-35}$  on the bottom.
13. **Your Turn — Same steps, your numbers. Boards up when done.** Catches the un-raised coefficient ( $x^6y^6/3$ ) and the b-that-did-not-move ( $x^6/(27y^6)$ ).
14. **Your Turn — Answer.** Reveal; name what a wrong board most likely did (see the note above).
- 15-16. **Board 1.**  $x^{-8} \rightarrow \frac{1}{x^8}$   $-x^8$  — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents];  $-8x$  — exponent multiplied into the base
- 17-18. **Board 2.**  $\frac{1}{y^{-9}} \rightarrow y^9$   $1/y^9$  — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]
- 19-20. **Board 3.** The brightness of a lamp from  $x$  meters away is  $5x^{-2}$  units. The lamp is 15 centimeters tall. How bright is the lamp from 3 meters away?  $\rightarrow \frac{5}{9}$  The height of 15 centimeters is not needed.  $1/45$  — the coefficient was moved down with the variable [8.AR.1.1: a negative exponent applies only to its own base];  $45$  — the minus on the exponent was dropped,  $5 \cdot 3^2$ ;  $-45$  — negative exponent read as a negative value [B1G-M 8.NSO.1.3:  $-b$  vs  $b^{-1}$ ]; anything built from 15 — the unneeded figure was used
- 21-22. **Board 4.**  $p^{-7}q^{-2} \rightarrow \frac{1}{p^7q^2}$   $1/(pq)^9$  — exponents on different bases were added [8.AR.1.1: like bases only];  $-p^7q^2$  — read as a negative
- 23-24. **Board 5.**  $x^3 \cdot x^{-14}$  Which expression is equivalent?  $\rightarrow \mathbf{C - 1/x^{11}}$  (A)  $x^{11}$  read as the opposite of  $x^{11}$  [B1G-M 8.AR.1.1 misconception: negative exponents]; (B)  $3 \cdot$

(-14) — multiplied the exponents [power of a power confused with a product]; (D) 3 – (-14) — subtracted instead of adding a negative [product law: add the exponents] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**25-26. Board 6.**  $(3x^{-5})^2$  Which expression is equivalent? → **D –  $9/x^{10}$**  (A)  $-5 + 2 = -3$  instead of  $-5 \cdot 2$  [power of a power confused with a product of powers]; (B) the coefficient was moved below the bar with the x [8.AR.1.1: a negative exponent applies only to its own base]; (C) the coefficient was not squared [8.AR.1.1: power of a product] Split (no option near two-thirds)? Sixty seconds, convince your neighbour, re-vote, then reveal.

**27-28. Board 7.**  $\frac{a^2}{a^{11}}$  →  **$\frac{1}{a^9}$**   $a^9 - 11 - 2$  instead of  $2 - 11$  [quotient law: top minus bottom];  $a^9$  — correct but not in the positive-exponent form asked for

**29-30. Board 8.** Find the value of  $n$  :  $x^{-5} \cdot x^n = x^2$  →  **$n = 7$**   $-3$  — added the given exponents [8.AR.1.1: product law, solve  $-5 + n = 2$ ];  $-7$  — sign error in the integer subtraction

**31-32. Board 9.** How are  $x^5$  and  $x^{-5}$  related? Explain, and give the value of each when  $x = 2$ . →  **$x^{-5}$  is the reciprocal of  $x^5$ . When  $x = 2$ :  $x^5 = 32$  and  $x^{-5} = 1/32$ .** ' $x^{-5} = -32$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]; ' $-10$ ' — exponent multiplied into the base

**33. Independent Set — Six questions. On your own, in writing.** Hand out the Independent Set; the same six questions are on the board.

**34. IXL — Last five minutes.** IXL: every listed skill is required, SmartScore 67.

## Misconceptions to Watch

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**Board 1.**  $-x^8$  — negative exponent read as a negative expression [B1G-M 8.AR.1.1 misconception: negative exponents]

**Board 2.**  $1/y^9$  — the sign was changed but the power was not moved [8.AR.1.1: law applied in one direction only]

**Board 3.**  $1/45$  — the coefficient was moved down with the variable [8.AR.1.1: a negative exponent applies only to its own base]

**Board 4.**  $1/(pq)^9$  — exponents on different bases were added [8.AR.1.1: like bases only]

**Board 5.**  $x^{-11}$  read as the opposite of  $x^{11}$  [B1G-M 8.AR.1.1 misconception: negative exponents]

**Board 6.**  $-5 + 2 = -3$  instead of  $-5 \cdot 2$  [power of a power confused with a product of powers]

**Board 7.**  $a^9 - 11 - 2$  instead of  $2 - 11$  [quotient law: top minus bottom]

**Board 8.**  $-3$  — added the given exponents [8.AR.1.1: product law, solve  $-5 + n = 2$ ]

**Board 9.** ' $x^{-5} = -32$ ' or 'they are opposites' — negative exponent read as a negative [B1G-M 8.AR.1.1 misconception: negative exponents]

**Materials** Whiteboards and markers. Calculators are allowed on everything;  $4^5 = 1,024$  in Example 2 is the one computation worth a calculator.